EXAM PAPERS PRACTICE

L3 Central tendency and variation

Name: ____________________

Class: ____________________

Date: ____________________

Time: **325 minutes**

Marks: **260 marks**

Comments:

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Q1.

Given that $\sum x = 364$, $\sum x^2 = 19412$, $n = 10$, find σ, the standard deviation of X.

Circle your answer.

24.8 44.1 616.2 1941.2

(Total 1 mark)

Q2.

A survey of 120 adults found that the volume, X litres per person, of carbonated drinks they consumed in a week had the following results:

$$\sum x = 165.6 \qquad \sum x^2 = 261.8$$

(a) (i) Calculate the mean of X.

(1)

(ii) Calculate the standard deviation of X.

(2)

(b) Assuming that X can be modelled by a normal distribution find

(i) $P(0.5 < X < 1.5)$

(2)

(ii) $P(X = 1)$

(1)

(c) Determine with a reason, whether a normal distribution is suitable to model this data.

(2)

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(d) It is known that the volume, Y litres per person, of energy drinks consumed in a week may be modelled by a normal distribution with standard deviation 0.21

Given that $P(Y > 0.75) = 0.10$, find the value of μ, correct to three significant figures.

(4)

(Total 12 marks)

Q3.

The data in the table gives the average amount, in millilitres, consumed per person per week of dried milk products for the East (E) and for the North West (NW) regions.

The line graph illustrates these data.

	Year since 2000													
	1	2	3	4	5	6	7	8	9	10	11	12	13	14

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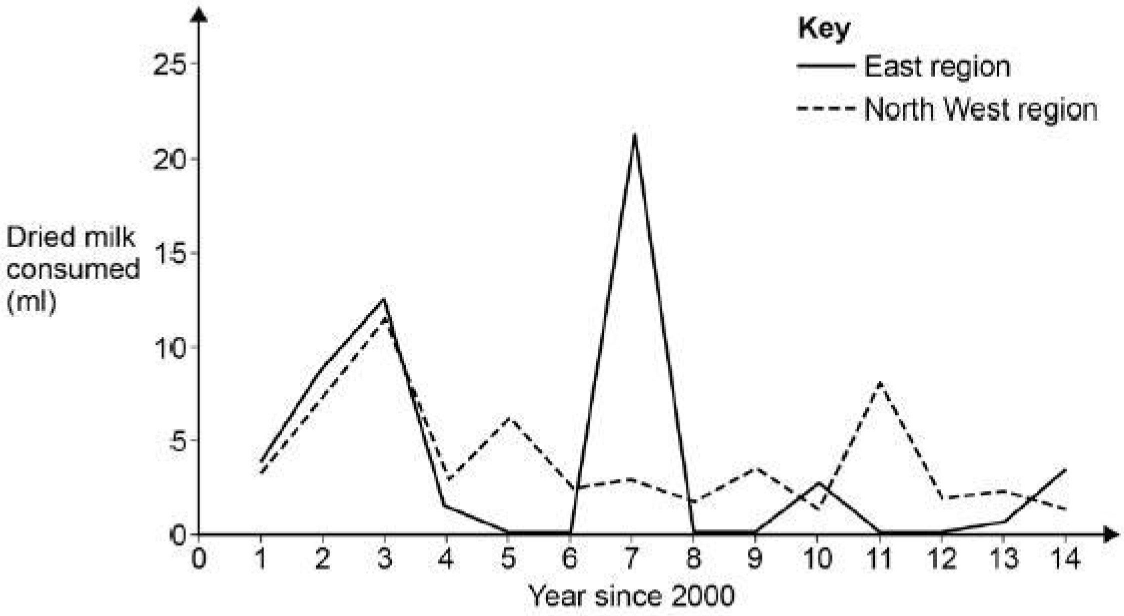
E	4	9	12	1	0	0	21	0	0	3	0	0	1	0
NW	3	7	12	3	6	2	3	2	3	1	8	2	2	1

Line graph to illustrate average dried milk consumption for E and NW regions



(a) (i) Identify, by year and region, a likely outlier.

(1)

(ii) A value in a data set can be identified as an outlier if it exceeds Q_3 + 1.5 × IQR
where IQR is the Interquartile range.

Finding any necessary statistics from the given data, confirm that the suggested outlier in part **(a)(i)** can be confirmed as an outlier.

(3)

(b) For the region with the confirmed outlier, explain why the mean is **not** an appropriate measure of average to use.

(1)

(c) Suggest a reason for the occurrence of the identified outlier.

(1)

(Total 6 marks)

Q4.

In the South West region of England, 100 households were randomly selected and, for each household, the weekly expenditure, £X , per person on food and drink was recorded.

The maximum amount recorded was £40.48 and the minimum amount recorded was £22.00

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The results are summarised below, where x̄ denotes the sample mean.

$$\sum x = 3046.14 \qquad \sum (x-\bar{x})^2 = 1746.29$$

(a) (i) Find the mean of X

Find the standard deviation of X

(2)

(ii) Using your results from part **(a)(i)** and other information given, explain why the normal distribution can be used to model X.

(2)

(iii) Find the probability that a household in the South West spends less than £25.00 on food and drink per person per week.

(1)

(b) For households in the North West of England, the weekly expenditure, £Y, per person on food and drink can be modelled by a normal distribution with mean £29.55

It is known that P(Y < 30) = 0.55

Find the standard deviation of Y, giving your answer to one decimal place.

(3)

(Total 8 marks)

Q5.

The average maximum monthly temperatures, u degrees Fahrenheit, and the average minimum monthly temperatures, v degrees Fahrenheit, in New York City are as follows.

	Jan	**Feb**	**Mar**	**Apr**	**May**	**Jun**	**Jul**	**Aug**	**Sep**	**Oct**	**Nov**	**Dec**
Maximum (u)	39	40	48	61	71	81	85	83	77	67	54	41
Minimum (v)	26	27	34	44	53	63	68	66	60	51	41	30

Calculate, to one decimal place, the mean and the standard deviation of the 12 values of the average **maximum** monthly temperature.

(Total 2 marks)

Q6.

A sawmill's machine cuts stakes for use in constructing agricultural fences. The lengths of the cut stakes may be modelled approximately by a normal distribution.

The machine is set to cut stakes of length 1.5 m and is known to cut stakes to within 10 cm of any set length.

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Elmer, the sawmill's foreman, asked Ashley, a trainee office junior, to calculate the mean and the standard deviation for the lengths of a random sample of 10 such stakes selected from the machine's output.

Ashley reported back to Elmer that the mean length of his 10 selected stakes was 151.5 cm and that the standard deviation of their lengths was 26.6 cm.

Advise Elmer on whether **each** of Ashley's values could be correct. Give numerical support for your answers.

(4)

(Total 4 marks)

Q7.

Steve, an employee at HFM gentlemen's hairdressers, was thinking of purchasing the business from the current owner, Keith.

Prior to deciding whether or not to purchase the business, Steve counted the daily number of customers, x, for a period of 25 days, with the following results.

x	18	20	23	24	28	30	32	35	36	38	**Total**
f	1	1	1	3	6	3	5	3	1	1	25

(a) Calculate the mean number of customers per day.

(2)

(b) Steve also noted that, of the above customers, 219 were children who paid £5.50 each, 73 were seniors who paid £6.50 each and 438 were adults who paid £9.50 each.

Calculate, in £, the mean takings per day.

(2)

(c) Steve estimated that, in order to make a success of the business, he would need an average of more than 28 customers per day **and** average takings of at least £250 per day.

Using your answers from parts (a) and (b), advise Steve as to whether or not he should purchase the business. Give numerical justification for your answer.

(2)

(Total 6 marks)

Q8.

Katy works as a clerical assistant for a small company. Each morning, she collects the company's post from a secure box in the nearby Royal Mail sorting office.

Katy's supervisor asks her to keep a daily record of the number of letters that she collects.

Her records for a period of 175 days are summarised in the table.

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Daily number of letters (x)	Number of days (f)
0-9	5
10-19	16
20	23
21	27
22	31
23	34
24	16
25-29	10
30-34	5
35-39	3
40-49	4
50 or more	1
Total	175

(a) For these data:

(i) state the modal value;

(1)

(ii) determine values for the median and the interquartile range.

(3)

(b) The most letters that Katy collected on any of the 175 days was 54. Calculate estimates of the mean and the standard deviation of the daily number of letters collected by Katy.

(4)

(c) During the same period, a total of 280 letters was also delivered to the company by private courier firms.

Calculate an estimate of the mean daily number of **all** letters received by the company during the 175 days.

(2)

(Total 10 marks)

Q9.

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The number of matches in each of a sample of 85 boxes is summarised in the table.

Number of matches	Number of boxes
Less than 239	1
239–243	1
244–246	2
247	3
248	4
249	6
250	10
251	13
252	16
253	20
254	5
255–259	3
More than 259	1
Total	**85**

(a) For these data:

(i) state the modal value;

(1)

(ii) determine values for the median and the interquartile range.

(3)

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(b) Given that, on investigation, the 2 extreme values in the above table are 227 and 271, calculate estimates of the mean and the standard deviation.

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(4)

(c) For the numbers of matches in the 85 boxes, suggest, with a reason, the most appropriate measure of spread.

(2)

(Total 10 marks)

Q10.

The number of matches in each of a sample of 85 boxes is summarised in the table.

Number of matches	Number of boxes
Less then 239	1
239–243	1
244–246	2

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247	3
248	4
249	6
250	10
251	13
252	16
253	20
254	5
255–259	3
More then 259	1
Total	**85**

(a) For these data:

(i) state the modal value;

(1)

(ii) determine values for the median and the interquartile range.

(3)

(b) Given that, on investigation, the 2 extreme values in the above table are 227 and 271:

(i) calculate the range;

(1)

(ii) calculate estimates of the mean and the standard deviation.

(4)

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(c) For the numbers of matches in the 85 boxes, suggest, with a reason, the most appropriate measure of spread.

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(2)

(Total 11 marks)

Q11.

Lizzie, the receptionist at a dental practice, was asked to keep a weekly record of the number of patients who failed to turn up for an appointment. Her records for the first 15 weeks were as follows.

20 26 32 a 37 14 27 34 15 18 b 25 37 29 25

Unfortunately, Lizzie forgot to record the actual values for two of the 15 weeks, so she recorded them as a and b. However, she did remember that a < 10 and that b > 40.

(a) Calculate the median and the interquartile range of these 15 values.

(4)

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(b) Give a reason why, for these data:

(i) the mode is **not** an appropriate measure of average;

(ii) the standard deviation **cannot** be used as a measure of spread.

(2)
(Total 6 marks)

Q12.

Lizzie, the receptionist at a dental practice, was asked to keep a weekly record of the number of patients who failed to turn up for an appointment. Her records for the first 15 weeks were as follows.

20 26 32 a 37 14 27 34 15 18 b 25 37 29 25

Unfortunately, Lizzie forgot to record the actual values for two of the 15 weeks, so she recorded them as a and b. However, she did remember that $a < 10$ and that $b > 40$.

(a) Calculate the median and the interquartile range of these 15 values.

(4)

(b) Give a reason why, for these data:

(i) the mode is **not** an appropriate measure of average;

(ii) the standard deviation **cannot** be used as a measure of spread.

(2)

(c) Subsequent investigations revealed that the missing values were 8 and 43.

Calculate the mean and the standard deviation of the 15 values.

(2)
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(Total 8 marks)
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Q13.

Ms N Parker always reads the columns of announcements in her local weekly newspaper. During each week of 2008, she notes the number of births announced. Her results are summarised in the table.

Number of births	1	2	3	4	5	6	7	8
Number of weeks	1	2	9	13	7	13	6	1

(a) Calculate the mean, median and modes of these data.

(5)

(b) State, with a reason, which of the three measures of average in part (a) you consider to be the **least** appropriate for summarising the number of births.

(2)
(Total 7 marks)

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Q14.

A survey of all the households on an estate is undertaken to provide information on the number of children per household.

The results, for the 99 households with children, are shown in the table.

Number of children (x)	1	2	3	4	5	6	7
Number of households (f)	14	35	25	13	9	2	1

(a) For these 99 households, calculate values for the mean and the standard deviation.

(3)

(b) In fact, 163 households were surveyed, of which 64 contained no children.

(i) For all 163 households, calculate the value for the mean number of children per household.

(2)

(ii) State whether the value for the standard deviation, when calculated for all 163 households, will be smaller than, the same as, or greater than that calculated in part (a).

(1)

(iii) It is claimed that, for all 163 households on the estate, the number of children per household may be modelled approximately by a normal distribution.

Comment, with justification, on this claim. Your comment should refer to a fact other than the discrete nature of the data.

(2)

(Total 8 marks)

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Q15.

A survey of all the households on an estate is undertaken to provide information on the number of children per household.

The results, for the 99 households with children, are shown in the table.

Number of children (x)	1	2	3	4	5	6	7
Number of households (f)	14	35	25	13	9	2	1

(a) For these 99 households, calculate values for:

(i) the median and the interquartile range;

(3)

(ii) the mean and the standard deviation.

(3)

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(b) In fact, 163 households were surveyed, of which 64 contained no children.

(i) For all 163 households, calculate the value for the mean number of children per household.

(2)

(ii) State whether the value for the standard deviation, when calculated for all 163 households, will be smaller than, the same as, or greater than that calculated in part (a) (ii).

(1)

(iii) It is claimed that, for all 163 households on the estate, the number of children per household may be modelled approximately by a normal distribution.

Comment, with justification, on this claim. Your comment should refer to a fact other than the discrete nature of the data.

(2)

(Total 11 marks)

Q16.

For each of the Premiership football seasons 2004/05 and 2005/06, a count is made of the number of goals scored in each of the 380 matches. The results are shown in the table.

Number of goals scored in a match	**Number of matches**	
	2004/05	**2005/06**
0	30	32
1	79	82
2	99	95
3	68	78
4	60	48
5	24	30
6	11	9
7	6	6
8	2	0
9	1	0
Total	**380**	**380**

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(a) For the number of goals scored in a match during the **2004/05** season:

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(i) determine the median and the interquartile range;

(4)

(ii) calculate the mean and the standard deviation.

(4)

(b) Two statistics students, Jole and Katie, independently analyse the data on the number of goals scored in a match during the **2005/06** season.

- Jole determines correctly that the median is 2 and that the interquartile range is also 2.
- Katie calculates correctly, to two decimal places, that the mean is 2.48 and that the standard deviation is 1.59.

Use your answers from part (a), together with Jole's and Katie's results, to compare briefly the two seasons with regard to the average and the spread of the number of goals scored in a match.

(2)

(Total 10 marks)

Q17.

For each of the Premiership football seasons 2004/05 and 2005/06, a count is made of the number of goals scored in each of the 380 matches. The results are shown in the table.

Number of goals scored in a match	**Number of matches**	
	2004/05	**2005/06**
0	30	32
1	79	82
2	99	95
3	68	78
4	60	48
5	24	30
6	11	9
7	6	6
8	2	0
9	1	0
Total	**380**	**380**

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(a) For the number of goals scored in a match during the **2004/05** season:

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(i) determine the median and the interquartile range;

(4)

(ii) calculate the mean and the standard deviation.

(4)

(b) Two statistics students, Jole and Katie, independently analyse the data on the number of goals scored in a match during the **2005/06** season.

- Jole determines correctly that the median is 2 and that the interquartile range is also 2.
- Katie calculates correctly, to two decimal places, that the mean is 2.48 and that the standard deviation is 1.59.

(i) Use your answers from part (a), together with Jole's and Katie's results, to compare briefly the two seasons with regard to the average and the spread of the number of goals scored in a match.

(2)

(ii) Jole claims that Katie's results must be wrong as 95% of values always lie within 2 standard deviations of the mean and (2.48 – 2 × 1.59) < 0 which is nonsense.

Explain why Jole's claim is incorrect. (You are not expected to confirm Katie's results.)

(2)
(Total 12 marks)

Q18.

The runs scored by a cricketer in 11 innings during the 2006 season were as follows.

47 63 0 28 40 51 a 77 0 13 35

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The exact value of a was unknown but it was greater than 100.

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(a) Calculate the median and the interquartile range of these 11 values.

(4)

(b) Give a reason why, for these 11 values:

(i) the mode is **not** an appropriate measure of average;

(ii) the range is **not** an appropriate measure of spread.

(2)
(Total 6 marks)

Q19.

The times, in seconds, taken by 20 people to solve a simple numerical puzzle were

17 19 22 26 28 31 34 36 38 39

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41 42 43 47 50 51 53 55 57 58

(a) Calculate the mean and the standard deviation of these times.

(3)

(b) In fact, 23 people solved the puzzle. However, 3 of them failed to solve it within the allotted time of 60 seconds.

Calculate the median and the interquartile range of the times taken by all 23 people.

(4)

(c) For the times taken by all 23 people, explain why:

(i) the mode is **not** an appropriate numerical measure;

(ii) the range is **not** an appropriate numerical measure.

(2)

(Total 9 marks)

Q20.

A library allows each member to have up to 15 books on loan at any one time.

The table shows the numbers of books currently on loan to a random sample of 95 members of the library.

Number of books on loan	0	1	2	3	4	5–9	10–14	15
Number of members	4	13	24	17	15	11	5	6

(a) For these data:

(i) state values for the mode and range;

(2)

(ii) determine values for the median and interquartile range;

(4)

(iii) calculate estimates of the mean and standard deviation.

(4)

(b) Making reference to your answers to part (a), give a reason for preferring:

(i) the median and interquartile range to the mean and standard deviation for summarising the given data;

(1)

(ii) the mean and standard deviation to the mode and range for summarising the given data.

(1)

(Total 12 marks)

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Q21.

The time, x seconds, spent by each of a random sample of 100 customers at an automatic teller machine (ATM) is recorded. The times are summarised in the table.

Time (seconds)	Number of customers
$20 < x \leq 30$	2
$30 < x \leq 40$	7
$40 < x \leq 60$	18
$60 < x \leq 80$	27
$80 < x \leq 100$	23
$100 < x \leq 120$	13
$120 < x \leq 150$	7
$150 < x \leq 180$	3
Total	**100**

Calculate estimates for the mean and standard deviation of the time spent at the ATM by a customer.

(Total 4 marks)

Q22.

Kirk and Les regularly play each other at darts.

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(a) The probability that Kirk wins any game is 0.3, and the outcome of each game is independent of the outcome of every other game.

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Find the probability that, in a match of 15 games, Kirk wins:

(i) exactly 5 games;

(3)

(i) fewer than half of the games;

(3)

(ii) more than 2 but fewer than 7 games.

(3)

(b) Kirk attends darts coaching sessions for three months. He then claims that he has a probability of 0.4 of winning any game, and that the outcome of each game is independent of the outcome of every other game.

(i) Assuming this claim to be true, calculate the mean and standard deviation for

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the number of games won by Kirk in a match of 15 games.

(3)

(ii) To assess Kirk's claim, Les keeps a record of the number of games won by Kirk in a series of 10 matches, each of 15 games, with the following results:

8 5 6 3 9 12 4 2 6 5

Calculate the mean and standard deviation of these values.

(2)

(iii) Hence comment on the validity of Kirk's claim.

(3)
(Total 17 marks)

Q23.

Katrina receives e-mail messages. The table below shows, for a random sample of 40 weekdays, the number of e-mail messages received by Katrina.

Number of e-mail messages	0	1	2	3	4	5	6	7	8
Number of weekdays	2	3	5	6	11	7	3	2	1

Calculate estimates for the mean and the standard deviation of the number of e-mail messages received per weekday by Katrina.

(Total 3 marks)

Q24.

On arrival at a business centre, all visitors are required to register at the reception desk. An analysis of the register, for a random sample of 100 days, results in the following information on the number, X, of visitors per day.

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Number of visitors per day	**Number of days**
1– 10	13
11– 20	33
21– 25	17
26– 30	12
31– 35	8
36– 40	5
41– 50	5

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51– 100	7
Total	**100**

(a) Calculate an estimate of:

(i) μ, the mean number of visitors per day;

(ii) σ, the standard deviation of the number of visitors per day.

(4)

(b) Give a reason, based upon the data provided, why X is **unlikely** to be normally distributed.

(1)

(c) (i) Give a reason why $\overline{X}$, the mean of a random sample of 100 observations on X, may be assumed to be normally distributed.

(1)

(ii) State, in terms of μ and σ, the mean and variance of $\overline{X}$.

(2)

(d) Hence construct a 99% confidence interval for μ.

(4)

(e) The receptionist claims that she registers on average more than 30 visitors per day, and frequently registers more than 50 visitors on any one day.

Comment on **each** of these **two** claims.

(4)
(Total 16 marks)

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Q25.

The heights, x cm, of the 10 girls in an athletics team are such that

$$\Sigma x = 1550 \quad \text{and} \quad \Sigma x^2 = 241\,250$$

(a) Find the mean and variance of x.

(3)

(b) One girl, who is 173 cm tall, drops out of the team. Calculate the mean and variance of the heights of the remaining nine members of the team.

(5)
(Total 8 marks)

Q26.

A survey of 100 customers' waiting times in a post office shows the following results.

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Time in minutes	0 – 1	1 – 2	2 – 3	3 – 6	6 –
Frequency	38	32	24	6	0

Calculate estimates of the mean and standard deviation of the waiting times.

(Total 6 marks)

Q27.

A student used the internet to obtain 40 records of the maximum daily temperature, x °F, over a 40-day period in 1900. The results were summarised as

$$\Sigma x = 2160, \qquad \Sigma x^2 = 116\,680.$$

(a) Find the mean and standard deviation of these temperatures.

(3)

To enable the data to be compared with results for the year 2000, the student converted each temperature to y °C by means of the linear scaling

$$y = \frac{5}{9}x - \frac{160}{9}$$

(b) Find the mean and standard deviation of the temperatures in °C.

(3)

(Total 6 marks)

Q28.

A chef prepares 10 meals using portions of meat bought in a shop.

The masses, x grams, of the 10 portions of meat are such that

$$\Sigma x = 2000 \quad \text{and} \quad \Sigma x^2 = 409\,000.$$

(a) Find the mean and standard deviation of these masses.

(4)

(b) Deduce the mean and standard deviation of the costs of the portions of meat, given that the meat costs 2 pence per gram.

(2)

(c) Each meal costs £1 in addition to the cost of the meat. Write down the mean and standard deviation of the total cost of a meal.

(2)

(Total 8 marks)

Q29.

Each house in a local authority area is supplied with a recycling box for discarded bottles. The council arranges a fortnightly collection of the boxes from outside the houses.

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In a street containing 30 houses the number of boxes placed outside for a particular collection is a random variable X. The values of X for 9 consecutive collections were:

15 12 10 11 12 14 11 10 13

(a) Calculate the mean and standard deviation of the given data.

(2)

(b) Use the data to estimate p, the proportion of houses that a box will be placed outside for a particular collection.

(1)

(c) Suggest the name of a distribution which might provide a suitable model for the random variable X.

(1)

(d) Use the distribution you have suggested in part (c) and the value of p you have estimated in part (b) to find:

(i) the probability that X exceeds 15;

(3)

(ii) the mean and standard deviation of X.

(3)

(e) By comparing your answers to part (a) with your answers to part (d)(ii), explain whether the distribution you have suggested in part (c) does provide a suitable model for X.

(2)

(f) Regardless of your calculations and of your answer to part (e), give a reason why the distribution you have suggested in part (c) may not provide a suitable model for X.

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(1)

(Total 13 marks)

Q30.

Siballi is a student who travels to and from university by bus. He believes that the probability of having to wait more than 6 minutes to catch a bus is 0.4 and is independent of the time of day and direction of travel.

(a) Assuming that Siballi's beliefs are correct, find the probability that, during a particular week when he catches 10 buses, the number of times he has to wait more than 6 minutes is:

(i) 3 or fewer;

(3)

(ii) more than 4.

(2)

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(b) Assuming that Siballi's beliefs are correct, calculate values for the mean and standard deviation of the number of times he has to wait more than six minutes to catch a bus in a week when he catches 10 buses.

(3)

(c) During a thirteen-week period, the numbers of times (out of 10) he had to wait more than six minutes to catch a bus were as follows:

4 8 8 9 3 2 2 7 0 1 5 2 0

(i) Calculate the mean and standard deviation of these data.

(2)

(ii) State, giving reasons, whether your answers to part (c)(i) support Siballi's beliefs that the probability of having to wait more than 6 minutes to catch a bus is 0.4 and is independent of the time of day and direction of travel.

(3)

(Total 13 marks)

Q31.

A part of a transport study, a sample of people who commuted regularly from a town in Surrey into London was asked to complete a questionnaire.

(a) The third question asked for an estimate of the time taken on their most recent journey into London. The replies are summarised below.

Time (minutes)	Frequency
35 –	12
55 –	54
75 –	68
95 –	41
115 – 155	23

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Calculate estimates of the mean and the standard deviation of these times.

(5)

(b) A sample of people who commuted regularly from a town in Essex into London was also asked the third question. Their answers had a mean of 86 minutes with a standard deviation of 35 minutes

Compare briefly, the journey times of commuters from the two towns.

(2)

(Total 7 marks)